MPG No. 8.1.24 mmm [Type I, orthorhombic] [T tensor]

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & T_{yy} & 0 \\ 0 & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$T_{yy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$
$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 \\ T_{xxyy} & T_{yyyy} & T_{yyzz} & 0 & 0 & 0 \\ T_{xxzz} & T_{yyzz} & T_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)}$$
$$T_{xxzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$
$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$
$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$
$$T_{zxzx} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$
$$T_{xyxy} = \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 \\ -T_{xxyy} & 0 & T_{yyzz} & 0 & 0 & 0 \\ -T_{xxzz} & -T_{yyzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxyy} = 4\mathbb{M}_{xyz}^{(1)} - 2\mathbb{T}_v^{(3)}$$
$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(3)}$$
$$T_{yyzz} = 4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(3)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & 0 \\ 0 & T_{zxzx} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(4)}$$
$$T_{zxzx} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 2\mathbb{T}_v^{(4)}$$
$$T_{xyxy} = \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzyz} & 0 & 0 \\ 0 & T_{zxzx} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)}$$
$$T_{zxzx} = \mathbb{M}_{xyz}^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)}$$
$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + 2\mathbb{T}_v^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(3)} - 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)}$$
$$T_{zxzx} = \mathbb{M}_{xyz}^{(3)} + 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)}$$
$$T_{xyxy} = \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_v^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yyxx} & T_{yyyy} & T_{yyzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{zzxx} & T_{zzyy} & T_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxzx} & 0 & 0 & T_{zxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyyx} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$
$$T_{xxyy} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)}$$
$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$
$$T_{yyxx} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} + 2\mathbb{T}_v^{(3)}$$
$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$
$$T_{yyzz} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$
$$T_{zzxx} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$
$$T_{zzyy} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$
$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$
$$T_{yzzy} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$
$$T_{zxzx} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$
$$T_{zxxz} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$
$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(5)}$$
$$T_{xyyx} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxzx} & 0 & 0 & T_{zxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= \mathbb{M}^{(3)}_{xyz} + \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} - 3\mathbb{T}^{(6)}_{u} + 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{yzzy} &= \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(3)}_{0} + 2\mathbb{T}^{(4)}_{u} - 3\mathbb{T}^{(6)}_{u} - 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{zxzx} &= \mathbb{M}^{(3)}_{xyz} + \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} + 3\mathbb{T}^{(6)}_{u} - 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{zxxz} &= \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(3)}_{0} + 2\mathbb{T}^{(4)}_{u} + 3\mathbb{T}^{(6)}_{u} + 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{xyxy} &= \mathbb{M}^{(3)}_{xyz} + \mathbb{T}^{(3)}_{0} + 4\mathbb{T}^{(4)}_{u} + 2\mathbb{T}^{(6)}_{v} \\
T_{xyyx} &= \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(3)}_{0} - 4\mathbb{T}^{(4)}_{u} + 2\mathbb{T}^{(6)}_{v}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & 0 & 0 \\ 0 & T_{yyyy} & 0 \\ 0 & 0 & T_{zzzz} \\ 0 & 0 & T_{yyzz} \\ T_{zzxx} & 0 & 0 \\ 0 & T_{xxyy} & 0 \\ 0 & T_{yzzy} & 0 \\ 0 & 0 & T_{zxxz} \\ T_{xxyy} + T_{yyzz} - T_{yzzy} - T_{zxxz} + T_{zzxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= 3\mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{4} - 2\mathbb{T}^{(1)}_{u} - 4\mathbb{T}^{(2)}_{u} + 2\mathbb{T}^{(1)}_{v} + 4\mathbb{T}^{(2)}_{v} \\
T_{yyyy} &= 3\mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} + \mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{4} - 2\mathbb{T}^{(1)}_{u} - 4\mathbb{T}^{(2)}_{u} - 2\mathbb{T}^{(1)}_{v} - 4\mathbb{T}^{(2)}_{v} \\
T_{zzzz} &= 3\mathbb{T}^{(1)}_{0} + 2\mathbb{T}^{(1)}_{4u} + 2\mathbb{T}^{(1)}_{4} + 4\mathbb{T}^{(1)}_{u} + 8\mathbb{T}^{(2)}_{u} \\
T_{yyzz} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(1)}_{u} + 3\mathbb{T}^{(3)}_{u} - \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v} \\
T_{zzxx} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} + \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(1)}_{u} - 3\mathbb{T}^{(3)}_{u} + \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v} \\
T_{xxyy} &= \mathbb{T}^{(1)}_{0} + 2\mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4} - 2\mathbb{T}^{(1)}_{u} - 2\mathbb{T}^{(3)}_{v} \\
T_{yzzy} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(2)}_{u} + 3\mathbb{T}^{(5)}_{u} - \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v} \\
T_{zxxz} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} + \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(2)}_{u} - 3\mathbb{T}^{(5)}_{u} + \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v}
\end{aligned}$$